

MANITOWOC COUNTY AP, WI, USA

WMO: 726455

Lat: 44.133N

Lon: 87.667W

Elev: 198

StdP: 98.96

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94897

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.5	-17.8	-26.8	0.3	-19.1	-23.4	0.5	-17.2	11.8	-7.6	10.9	-6.3	5.1	280	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	29.5	22.1	27.8	21.4	26.6	20.5	23.8	28.2	22.7	26.6	21.6	25.2	4.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.5	26.8	21.3	16.4	25.3	20.4	15.4	24.2	72.3	28.4	67.5	26.4	63.8	25.3	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.2	8.3	DB	-23.3	32.8	3.4	2.3	-25.8	34.5	-27.8	35.8	-29.7	37.1	-32.2	38.8	(4)
(5)			WB	-23.8	25.7	2.9	1.1	-25.9	26.5	-27.6	27.1	-29.2	27.8	-31.3	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-6.5	-5.2	-0.1	5.9	11.8	17.5	20.6	19.8	16.0	9.5	2.9	-3.5
	DBStd	10.57	6.37	6.03	5.47	4.18	4.05	3.72	3.18	3.02	3.90	4.35	4.81	5.86
	HDD10.0	2114	512	426	314	135	26	1	0	0	2	63	218	418
	HDD18.3	4173	771	659	570	373	206	58	14	18	89	274	464	676
	CDD10.0	1189	0	0	2	12	83	226	328	304	183	48	3	0
	CDD18.3	207	0	0	0	0	3	34	83	64	21	2	0	0
	CDH23.3	1359	0	0	1	5	44	263	592	343	102	9	0	0
	CDH26.7	309	0	0	0	1	7	60	158	65	17	1	0	0
(14) Wind	WSAvg	3.9	4.6	4.6	4.3	4.3	3.8	3.2	3.2	3.1	3.5	4.0	4.4	4.3
(15) Precipitation	PrecAvg	742	41	30	46	70	77	82	81	85	78	57	58	42
	PrecMax	1132	260	71	132	194	220	217	206	241	319	142	158	104
	PrecMin	528	2	0	1	14	15	10	21	4	4	2	2	2
	PrecStd	128	40	19	32	35	43	45	39	47	55	29	36	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	9.2	17.6	22.9	27.5	31.1	32.9	31.2	29.0	24.1	17.7	11.3
		MCWB	4.9	5.8	13.4	15.4	19.1	23.1	24.5	23.7	22.2	18.1	12.6	9.5
	2%	DB	5.0	6.2	12.5	17.7	24.0	28.1	30.0	28.0	26.2	20.9	13.8	7.9
		MCWB	3.0	3.5	8.7	11.8	17.3	21.0	22.8	21.8	20.4	16.3	10.0	5.8
	5%	DB	3.0	3.9	9.7	14.5	21.7	26.3	28.0	27.0	23.9	18.0	11.5	6.1
		MCWB	1.8	2.2	6.6	9.2	15.7	19.8	21.9	21.1	19.2	14.5	8.6	4.2
	10%	DB	2.2	2.6	7.2	12.4	18.9	24.1	27.0	25.2	22.5	16.4	9.5	3.9
		MCWB	1.1	1.1	4.5	7.9	14.0	18.6	21.2	20.0	18.5	13.4	7.4	2.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	6.8	13.9	16.0	20.3	24.2	25.7	25.1	23.0	19.9	13.7	10.1
		MCDB	7.5	8.6	17.2	22.3	25.6	29.4	31.2	29.4	27.0	23.1	16.8	11.0
	2%	WB	3.4	4.0	9.4	12.7	18.4	22.2	24.2	23.3	21.5	17.2	10.9	6.5
		MCDB	4.5	6.0	12.0	17.1	22.9	26.9	28.7	27.1	24.9	19.2	12.9	7.5
	5%	WB	2.2	2.5	6.8	10.2	16.5	20.7	23.0	22.1	20.4	15.3	9.0	4.4
		MCDB	3.2	3.9	9.1	13.7	20.7	25.2	27.0	25.2	23.1	17.8	11.0	5.8
	10%	WB	0.8	1.3	4.8	8.2	14.6	19.4	21.9	21.1	19.3	13.6	7.6	2.7
		MCDB	2.0	2.6	7.1	11.6	18.3	23.4	25.6	24.1	22.0	16.1	9.3	4.0
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.6	7.7	8.5	9.6	9.7	9.5	8.7	9.2	8.1	7.1	6.7
		MCDBR	7.5	7.7	10.9	13.1	13.7	12.1	11.0	10.0	10.8	10.8	9.9	7.9
		MCWBR	6.6	6.3	8.1	8.5	8.5	7.0	6.1	5.7	6.7	7.4	7.6	6.7
	5% WB	MCDBR	7.1	7.2	10.6	12.4	12.6	11.3	10.1	8.9	9.6	9.7	9.0	7.3
		MCWBR	6.5	6.1	8.3	8.8	8.7	7.4	6.4	5.7	6.5	7.5	7.5	6.6
(40) Clear-Sky Solar Irradiance	taub		0.278	0.310	0.332	0.380	0.403	0.418	0.408	0.403	0.379	0.338	0.305	0.280
	taud		2.440	2.338	2.391	2.301	2.289	2.293	2.326	2.336	2.388	2.497	2.519	2.486
	Ebn at Noon		857	879	906	883	869	855	858	850	844	841	818	821
	Edh at Noon		72	95	104	123	129	129	124	118	104	82	67	63
(44) All-Sky Solar Radiation	RadAvg		1.48	2.45	3.62	4.51	5.41	5.95	6.10	5.21	4.13	2.60	1.62	1.15
	RadStd		0.19	0.26	0.34	0.43	0.44	0.42	0.27	0.34	0.33	0.27	0.15	0.12

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page